# Nisa Nur Çolak


**Control and Automation Engineering Student**

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## SUMMARY


Control and Automation Engineering student at Yıldız Technical University with a strong foundation in system dynamics, mathematical modeling, and control system design. Expanding technical capabilities through the integration of data-driven methods and artificial intelligence into control-oriented engineering approaches. Interested in contributing to advanced control and simulation-based engineering solutions for autonomous and defense-related systems within multidisciplinary R&D environments.


## EXPERIENCE

**Vehicle Control Unit (VCU) Support Team Member | YTU Racing** — **Nov 2024 - Sep 2025**

- Conducted research on signal conditioning and sensor fusion for the Vehicle Control Unit to ensure high-fidelity data acquisition and enhance control system reliability.
- Analyzed vehicle dynamics and communication architectures (CAN-bus) to support technical reporting and system design workflows.
- Collaborated in a multidisciplinary team environment, contributing to technical reporting and system design support

**YTU ASIL (Advanced Systems and Invention Laboratories)**

- Selected as an AI Team candidate, undergoing intensive technical training in Supervised and Unsupervised Learning, Reinforcement Learning, and Machine Learning architectures.

## PROJECTS

**TÜBİTAK 2209-A Research Project (Ongoing)**

Project: In-Context Learning Based Meta-Filter Design for Autonomous Platooning

- Investigating state estimation challenges and stability criteria in autonomous platooning systems.
- Developing a data-driven meta-filter utilizing Transformer architectures and In-Context Learning to address non-linear state estimation challenges in autonomous platooning.

## SKILLS

**CONTROL & MODELING**

- Control System Design
- System Dynamics
- Mathematical Modeling
- Stability Analysis.

**SIMULATION&SYSTEMS**

- MATLAB & Simulink (High-fidelity system modeling, control loops, and performance simulations).

**SOFTWARE & TOOLS**

- C (Algorithmic Logic Design)
- PLC Programming (Omron CX-Programmer, PC Worx)
- MS Office.

**AI & DECISION SYSTEMS**

- Python (NumPy, Pandas, Scikit-Learn), PyTorch (Basic)
- Supervised & Unsupervised Learning, Machine Learning Fundamentals.

**AREAS OF INTEREST**

- Autonomous Systems, Decision Support Systems, Sensor Fusion, Data-Driven Control.

## EDUCATION

**B.Sc. in Control and Automation Engineering (English)**
**Yıldız Technical University (YTU) — Istanbul, Turkiye**

## ADDITIONAL INFORMATION

- **Languages:** English (C1), Turkish (Native)
- **Certifications:** Training Certificate in Autonomous Driving Technologies - National Technology Academy (2026) , Supervised Machine Learning: Regression and Classification (2026), Google AI Essentials (2025), BTK Academy C Programming Language Training Certificate (2025), MathWorks MATLAB Onramp Course Completion (2024), McKinsey.org Forward Program (2025), Esmiyor x Schneider Electric Ideathon Participation Certificate (2024)